

Crispy Fried Chicken Theory of Gravity
A Covariant Aether-Scalar-Tensor Framework with an Exponential
MOND Constitutive Response*

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Abstract

We formulate a covariant modified-gravity construction in the aether-scalar-tensor (AeST) class in which the galactic modified-Newtonian-dynamics (MOND) response is fixed to the exponential interpolation function $\mu(y) = 1 - e^{-y}$. The proposal is built from a Lorentzian metric $g_{\mu\nu}$, a unit-timelike vector field A^μ , a scalar field ϕ , and a Lagrange multiplier enforcing $A^\mu A_\mu = -1$. The scalar-gradient invariant is taken to be $Y = (g^{\mu\nu} + A^\mu A^\nu)\nabla_\mu\phi\nabla_\nu\phi$, while $Q = A^\mu\nabla_\mu\phi$ controls the homogeneous cosmological sector. We choose the free function so that its derivative with respect to Y is exactly $1 - e^{-\sqrt{Y}/a_0(Q)}$. In the quasistatic weak-field regime this produces the nonlinear MOND Poisson operator $\nabla \cdot [\mu(|\nabla\phi|/a_0)\nabla\phi]$, and in lapse variables the corresponding constraint can be written as $D_i[(1 - e^{-y})c^2 D^i \ln N] = 4\pi G\rho_m$ with $y = (c^2/a_0)\sqrt{D_i \ln N D^i \ln N}$. The construction separates the degenerate $Y = 0$ MOND branch from the homogeneous cosmological dynamics rather than excluding either sector. We distinguish results inherited from the established AeST framework from claims that are specific to the present constitutive choice. In particular, the paper does not claim a completed global stability proof for the exact model function; instead it gives the action, field equations, limits, consistency conditions, and a concrete set of falsifiable closure tests required for such a claim.

I. INTRODUCTION

Modified Newtonian dynamics (MOND) was introduced as an empirical modification of the low-acceleration gravitational law and has remained phenomenologically successful on galactic scales. A persistent theoretical problem is to construct a relativistic theory that preserves the MOND limit while remaining compatible with cosmology, gravitational lensing, and relativistic consistency conditions. Relativistic scalar-vector-tensor theories have therefore been explored as a route in which the MOND force is generated by additional gravitational fields while ordinary matter remains minimally coupled to a single physical metric.

The present work adopts the covariant aether-scalar-tensor (AeST) architecture of Skordis and Zlosnik and specializes its free scalar function to an exponential constitutive law. AeST

* The title is intentionally informal; all equations, assumptions, and tests in the body are stated in the conventional style of a theoretical-gravity manuscript.

was introduced as a relativistic MOND framework with a cosmological sector capable of mimicking the large-scale behavior usually attributed to cold dark matter, and subsequent analyses established its quasistatic phenomenology, linear stability around Minkowski space-time, and nonlinear Hamiltonian structure [1–4]. The full nonlinear Hamiltonian analysis finds six physical degrees of freedom in the generic theory; this is an important benchmark because the present proposal does not require the relativistic completion to propagate only the two tensor modes of general relativity [3].

The specific proposal here is motivated by a simple constitutive observation. If a MOND equation is written as

$$\nabla \cdot \left[\mu \left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho_b, \quad (1)$$

then the choice

$$\mu(y) = 1 - e^{-y} \quad (2)$$

provides the Newtonian asymptote $\mu \rightarrow 1$ for $y \gg 1$ and the deep-MOND asymptote $\mu \sim y$ for $y \ll 1$. The aim is to realize Eq. (2) as the derivative of a covariant action rather than inserting it directly at the level of the nonrelativistic field equation.

The resulting construction has two conceptually distinct sectors. The invariant Y governs spatial gradients relative to the aether and carries the MOND response, whereas Q controls the homogeneous background and can supply a cosmological effective fluid. This separation is useful because a homogeneous FLRW solution has $Y = 0$ while Q can remain nonzero. Thus the homogeneous branch does not require inversion of the MOND spatial operator.

The principal result is the covariant action and its exact constitutive derivative. The lapse-gradient equation quoted in phenomenological form is a weak-field reduction of this parent theory, not a replacement for the full relativistic action.

II. COVARIANT CONSTRUCTION

We work with metric signature $(-, +, +, +)$ and introduce a metric $g_{\mu\nu}$, scalar ϕ , unit-timelike vector A_μ , and Lagrange multiplier λ . Define

$$F_{\mu\nu} \equiv \nabla_\mu A_\nu - \nabla_\nu A_\mu, \quad (3)$$

$$J^\mu \equiv A^\nu \nabla_\nu A^\mu, \quad (4)$$

$$Q \equiv A^\mu \nabla_\mu \phi, \quad (5)$$

$$Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \phi \nabla_\nu \phi. \quad (6)$$

The tensor $P^{\mu\nu} = g^{\mu\nu} + A^\mu A^\nu$ is the projector onto the three-space orthogonal to A^μ , so $Y \geq 0$ on the unit-timelike branch.

The covariant action is taken to be

$$S = \frac{c^3}{16\pi G_*} \int d^4x \sqrt{-g} \left[R - 2\Lambda - \frac{K_B}{2} F_{\mu\nu} F^{\mu\nu} + (2 - K_B) (2J^\mu \nabla_\mu \phi - Y) - \mathcal{F}(Y, Q) - \lambda(A^\mu A_\mu + 1) \right] + S_m[g, \psi]. \quad (7)$$

This is the standard covariant AeST architecture, with the present model specified by the choice of the free function $\mathcal{F}(Y, Q)$ [1, 3]. Ordinary matter couples only to $g_{\mu\nu}$ through $S_m[g, \psi]$.

We write the free function as

$$\mathcal{F}(Y, Q) = (2 - K_B) \mathcal{J}(Y, Q) - 2\mathcal{K}(Q), \quad (8)$$

where $\mathcal{K}(Q)$ governs the homogeneous sector and $\mathcal{J}$ contains the MOND constitutive response.

Let

$$u \equiv \frac{\sqrt{Y}}{a_0(Q)}. \quad (9)$$

We define

$$\boxed{\mathcal{J}(Y, Q) = a_0^2(Q) [u^2 + 2(1 + u)e^{-u} - 2]} \quad (10)$$

Since $u^2 = Y/a_0^2(Q)$,

$$\begin{aligned} \frac{\partial \mathcal{J}}{\partial Y} &= \frac{1}{2u} \frac{d}{du} [u^2 + 2(1 + u)e^{-u} - 2] \\ &= 1 - e^{-u}. \end{aligned} \quad (11)$$

Hence the constitutive law is fixed exactly:

$$\boxed{\mathcal{J}_Y = 1 - e^{-\sqrt{Y}/a_0(Q)}}. \quad (12)$$

The homogeneous sector is represented by a function $\mathcal{K}(Q)$, following the convention $\mathcal{K}(Q) = -\mathcal{F}(0, Q)/2$. A useful phenomenological parameterization is an offset square-root form,

$$\mathcal{K}(Q) = -M^4 + \mu_Q^2 \Lambda_D^2 \left[1 - \sqrt{1 - \frac{(Q - Q_0)^2}{\Lambda_D^2}} \right], \quad (13)$$

combined with a pressure-linked acceleration scale

$$a_0^2(Q) = \kappa_a^2 G[-\mathcal{K}(Q)]. \quad (14)$$

Equation (14) is a model-building choice rather than a theorem. Its purpose is to make the MOND scale a property of the same field configuration that supplies the homogeneous cosmological sector.

III. FIELD EQUATIONS AND COVARIANT MATTER DYNAMICS

Varying Eq. (7) with respect to the metric yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \mathcal{H}_{\mu\nu} = \frac{8\pi G_*}{c^4} T_{\mu\nu}, \quad (15)$$

where $\mathcal{H}_{\mu\nu}$ contains the aether and scalar contributions. The detailed expression follows from the functional derivatives of $F_{\mu\nu}$, J^μ , Y , Q , and $\mathcal{F}$. The resulting structure is the covariant AeST stress tensor; its form and derivation are given explicitly in Refs. [1, 3].

Variation with respect to A_μ and ϕ gives, respectively,

$$\begin{aligned} 0 = & K_B \nabla_\mu F^{\mu\nu} + 2(2 - K_B) [\nabla_\mu \phi \nabla^\nu A^\mu - \nabla_\mu (A^\mu \nabla^\nu \phi)] \\ & - \lambda A^\nu - \frac{1}{2} \nabla^\nu \phi [2Q((2 - K_B) + \mathcal{F}_Y) + \mathcal{F}_Q], \end{aligned} \quad (16)$$

and

$$\begin{aligned} 0 = & \nabla_\mu (\mathcal{F}_Q A^\mu) - 2(2 - K_B) \nabla_\mu J^\mu \\ & + 2 \nabla_\mu [Q((2 - K_B) + \mathcal{F}_Y) A^\mu] + 2 \nabla_\mu [(2 - K_B) + \mathcal{F}_Y] \nabla^\mu \phi. \end{aligned} \quad (17)$$

These equations reduce to the familiar AeST field equations when the free function is left unspecified [8].

Because matter is minimally coupled to the metric, the diffeomorphism Ward identity gives

$$\boxed{\nabla_\mu T^{\mu\nu} = 0} \quad (18)$$

on-shell, provided the matter equations of motion and the full gravitational field equations hold. This is a central reason for retaining the covariant parent action rather than taking the lapse constraint itself as a fundamental noncovariant law.

IV. WEAK-FIELD AND QUASISTATIC LIMIT

Consider a weakly perturbed static configuration. Write the physical metric as

$$ds^2 = -(1 + 2\Phi/c^2)c^2 dt^2 + (1 - 2\Psi/c^2)\delta_{ij}dx^i dx^j + O(c^{-4}). \quad (19)$$

The aether provides a preferred timelike direction, while the scalar supplies the MOND sector. In the quasistatic regime the relevant spatial scalar invariant reduces to a positive function of the spatial scalar gradient. At the level of the constitutive operator,

$$Y \rightarrow |\nabla\phi|^2, \quad u \rightarrow \frac{|\nabla\phi|}{a_0}. \quad (20)$$

The scalar flux is therefore proportional to

$$\mathcal{J}_Y \nabla\phi = (1 - e^{-|\nabla\phi|/a_0}) \nabla\phi. \quad (21)$$

Consequently the static scalar equation has the MOND form

$$\boxed{\nabla \cdot [(1 - e^{-|\nabla\phi|/a_0}) \nabla\phi] = \mathcal{S}[\rho_b]}, \quad (22)$$

where the source normalization $\mathcal{S}$ is fixed by the precise AeST-to-Newtonian coupling convention. In a convention in which the physical Newtonian acceleration is directly identified with the total potential gradient, the target normalization is $4\pi G\rho_b$.

The two asymptotic regimes follow directly. For $|\nabla\phi| \gg a_0$,

$$\mu\left(\frac{|\nabla\phi|}{a_0}\right) = 1 + O(e^{-|\nabla\phi|/a_0}), \quad (23)$$

so Newtonian gravity is recovered exponentially fast. For $|\nabla\phi| \ll a_0$,

$$\mu\left(\frac{|\nabla\phi|}{a_0}\right) = \frac{|\nabla\phi|}{a_0} + O\left(\frac{|\nabla\phi|^2}{a_0^2}\right), \quad (24)$$

which yields the deep-MOND scaling.

V. LAPSE-GRADIENT FORM AND THE CENTRAL OBSERVABLE EQUATION

For a static metric with lapse N , the weak-field relation

$$N = 1 + \frac{\Psi}{c^2} + O(c^{-4}) \quad (25)$$

implies

$$c^2 D_i \ln N = D_i \Psi + O(c^{-2}). \quad (26)$$

It is therefore natural to define

$$y \equiv \frac{c^2}{a_0} \sqrt{D_i \ln N D^i \ln N}. \quad (27)$$

The phenomenological constraint can then be written compactly as

$$\boxed{D_i [(1 - e^{-y}) c^2 D^i \ln N] = 4\pi G \rho_m.} \quad (28)$$

Equation (28) is the central low-energy MOND constraint of the construction. In words, the gravitational field obeys a modified Poisson equation in which the response is $1 - e^{-y}$, with y measuring the local gravitational acceleration in units of a_0 .

The important interpretive point is that Eq. (28) is not being postulated as a standalone relativistic field theory. It is the quasistatic manifestation that the covariant constitutive sector is designed to reproduce. The relativistic completion is Eq. (7) together with Eqs. (8)-(13).

VI. THE $Y = 0$ AND HOMOGENEOUS BRANCHES

A global formulation must treat the homogeneous branch explicitly. On a spatially homogeneous FLRW background,

$$Y = 0, \quad Q = \dot{\phi} \quad (29)$$

when the cosmic aether is aligned with the comoving time direction. In this branch the MOND constitutive derivative satisfies

$$\mathcal{J}_Y \rightarrow 0, \quad (30)$$

while the Q sector remains dynamical through $\mathcal{K}(Q)$ and the background scalar equation.

The exponential law has the small-gradient expansion

$$1 - e^{-u} = u - \frac{u^2}{2} + O(u^3). \quad (31)$$

Thus, at $Y \rightarrow 0^+$, the spatial flux behaves as

$$\mathcal{J}_Y \nabla \phi \sim \frac{|\nabla \phi|}{a_0} \nabla \phi. \quad (32)$$

The principal coefficient of the spatial equation is therefore degenerate at exactly zero gradient. This should not be replaced by an artificial exclusion of the branch. Instead, the natural mathematical problem is a degenerate monotone elliptic equation. A complete global PDE theorem would require a proof of existence, uniqueness (modulo the appropriate gauge/boundary freedom), and continuous dependence through the $Y = 0$ set. The present paper identifies that as a mathematical completion condition rather than claiming it has already been proved.

The advantage of the covariant AeST parent is that the $Y = 0$ homogeneous branch does not imply that the entire theory loses its time evolution. The background equations depend on Q and $\mathcal{K}(Q)$ and therefore remain meaningful on FLRW. This is qualitatively different from a theory in which the only modification is an inverse spatial MOND operator.

VII. COSMOLOGY

For a flat FLRW metric,

$$ds^2 = -dt^2 + a^2(t) (dr^2 + r^2 d\Omega^2), \quad (33)$$

with $A_\mu = (-1, 0, 0, 0)$, one has $Y = 0$, $J^\mu = 0$, and $Q = \dot{\phi}$. The scalar equation reduces to a conserved quantity of the form

$$\frac{d}{dt} \mathcal{K}_Q + 3H \mathcal{K}_Q = 0, \quad (34)$$

so

$$\mathcal{K}_Q = \frac{I_0}{a^3}, \quad (35)$$

for integration constant I_0 . This is one reason AeST can generate an effective component with dust-like late-time behavior without placing an explicit pressureless matter field into the new sector [7, 8].

The Friedmann equations may be written schematically as

$$H^2 = \frac{8\pi G_*}{3}\rho + \rho_{\text{eff}}(Q, \mathcal{K}, \mathcal{K}_Q, \Lambda), \quad (36)$$

$$2\dot{H} + 3H^2 = -\frac{8\pi G_*}{c^2}p + p_{\text{eff}}(Q, \mathcal{K}, \mathcal{K}_Q, \Lambda), \quad (37)$$

with the precise coefficients depending on normalization conventions. Published AeST analyses derive the exact background equations for the general $\mathcal{F}(Y, Q)$ class and reconstruct choices of $\mathcal{K}(Q)$ capable of reproducing a Λ CDM-like expansion history [7, 8].

The present specialization makes a more restrictive statement: the same Q -dependent homogeneous sector is used to define the acceleration scale entering the MOND function. This provides a concrete mechanism by which the galactic scale can be associated with the cosmological state, but the mapping in Eq. (14) remains a hypothesis to be confronted with data.

VIII. DEGREES OF FREEDOM AND STABILITY

The nonlinear constraint analysis of the general AeST theory is an essential benchmark for the present construction. Bataki, Skordis, and Zlosnik find four first-class and four second-class constraints and six physical degrees of freedom at the fully nonlinear level [3]. The present model should therefore not be advertised as a two-degree-of-freedom modification of general relativity. Its additional physical sectors are part of the relativistic completion.

Linear perturbations around Minkowski spacetime have been analyzed for AeST and were found to contain massive propagating vector and scalar modes with parameter-dependent stability conditions; the published analysis also identifies a nonpropagating mode whose Hamiltonian behavior changes at a characteristic scale [2]. These results strongly motivate using the AeST Hamiltonian structure as the starting point for the present exact constitutive choice rather than re-counting only the tensor sector.

For the proposed theory, the required stability conditions are

$$Q_T > 0, \quad c_T^2 = 1, \quad (38)$$

for the tensor sector, together with positive kinetic matrices and nonnegative squared propagation speeds in all scalar and vector sectors on every background where the theory is claimed to apply. These are requirements, not assumptions; they must be checked for the specific $\mathcal{J}(Y, Q)$ and $\mathcal{K}(Q)$ adopted here.

IX. GRAVITATIONAL LENSING AND RELATIVISTIC POTENTIALS

Ordinary matter is minimally coupled to the physical metric. Therefore photons follow null geodesics of $g_{\mu\nu}$ and weak lensing is controlled by the combination of metric potentials appearing in

$$\frac{\Phi + \Psi}{c^2}. \quad (39)$$

The quasistatic AeST limit has been studied in spherical configurations and in weak lensing data. Those analyses show both the usefulness and the limitations of AeST: MOND-like behavior occurs in an intermediate regime, while additional scales can produce deviations at sufficiently large radius [4–6].

The present model therefore makes a sharp distinction between two statements. First, the exponential constitutive law fixes the scalar response to the form $1 - e^{-y}$. Second, the full lensing prediction follows only after the metric, aether, and scalar equations are solved together. The latter is a genuine relativistic prediction and must not be inferred from the scalar MOND equation alone.

X. RECOVERY OF GENERAL RELATIVITY

The high-acceleration limit of the constitutive law is

$$\lim_{y \rightarrow \infty} \mu(y) = 1. \quad (40)$$

Thus the MOND deformation of the static spatial response switches off exponentially as the acceleration increases. A complete relativistic GR limit additionally requires the aether-scalar stress tensor to become either negligible or observationally indistinguishable from a renormalization of the Einstein-Hilbert sector. This condition depends on the background values of Q , $a_0(Q)$, and the model parameters.

For Solar-System tests the relevant observables are the effective Newton constant, the post-Newtonian parameters γ and β , preferred-frame parameters, Shapiro delay, and perihelion precession. The requirement is not simply that $y \gg 1$, but that the full metric solution of Eq. (15) be close enough to the GR solution to satisfy experimental bounds.

XI. A GLOBAL CLOSURE PROGRAM

The construction presented here defines a concrete target for a global relativistic closure analysis. The required checks can be organized as follows:

1. derive the complete nonlinear Hamiltonian for the exact function $\mathcal{F}(Y, Q)$;
2. verify the global constraint rank, including homogeneous modes and the $Y = 0$ branch;
3. derive the complete scalar, vector, and tensor quadratic actions on FLRW;
4. prove positivity of the kinetic matrices and absence of gradient instabilities in the intended parameter region;
5. verify $c_T = 1$ in the exact model, not only in an undeformed reference theory;
6. solve the quasistatic equations beyond spherical symmetry and identify the transition to any AeST-specific far-field regime;
7. derive the full PPN limit and compare γ , β , and preferred-frame observables with experimental limits;
8. verify the cosmological background and linear perturbations against CMB and large-scale-structure data;
9. establish a mathematically controlled weak-solution formulation through $Y = 0$;
10. perform independent numerical red-team tests for extra branches, singular denominators, constraint-rank changes, ghosts, and superluminal characteristics.

Passing these tests would justify the label “globally closed” for the specified model. Failing any one of them would instead identify a precise obstruction. The distinction matters because the generic AeST Hamiltonian closure is already known, but the nonlinear function selected here is more restrictive and therefore requires its own perturbative and global analysis.

XII. DISCUSSION

The principal conceptual simplification of the construction is that the MOND response is encoded in a single constitutive derivative,

$$\boxed{\mu(u) = 1 - e^{-u}}, \quad u = \frac{\sqrt{Y}}{a_0(Q)}. \quad (41)$$

This function has the two asymptotes required by MOND while approaching the Newtonian response exponentially at high acceleration. Its primitive is elementary and therefore gives an explicit Lagrangian realization rather than a phenomenological interpolation inserted only after variation.

The second structural element is the separation of Y and Q . The spatial invariant Y controls the local MOND response, while Q permits a homogeneous cosmological sector even when the spatial MOND gradient vanishes. This avoids treating the $k = 0$ cosmological mode as an inverse problem for a degenerate MOND Laplacian.

The central observable statement can be compressed to

$$\boxed{D_i [(1 - e^{-y})c^2 D^i \ln N] = 4\pi G\rho_m, \quad y = \frac{c^2}{a_0} \sqrt{D_i \ln N D^i \ln N}.} \quad (42)$$

In plain language, the gravitational field obeys a modified Poisson equation in which the response changes from Newtonian to MOND according to the local acceleration measured in units of a_0 .

At present, the strongest defensible interpretation is that this equation is the low-energy signature of a covariant AeST realization with an explicitly chosen exponential MOND constitutive function. It is not yet a theorem that every global stability, strong-field, and cosmological requirement is simultaneously satisfied for every allowed parameter choice. Establishing or refuting that statement is precisely the purpose of the closure program above.

XIII. CONCLUSIONS

We have specified a covariant AeST-type action and a concrete free function for which the MOND constitutive derivative is exactly $1 - e^{-\sqrt{Y}/a_0(Q)}$. The construction yields the desired Newtonian and deep-MOND asymptotes and provides a covariant parent theory in which ordinary matter is minimally coupled to a single physical metric. The homogeneous FLRW

sector is carried by the independent invariant Q , allowing the $Y = 0$ branch to be treated as a legitimate background rather than excluded by construction.

The resulting proposal is therefore more complete than a standalone modified Poisson equation, but its scientific status remains that of a concrete relativistic model to be tested, not an already proven final theory. The decisive next steps are the global nonlinear constraint analysis for the exact function, stability of all perturbation sectors on cosmological backgrounds, a controlled treatment of the degenerate $Y = 0$ PDE branch, the full PPN limit, and quantitative lensing and structure-formation calculations. These are falsifiable calculations, not matters of nomenclature.

ACKNOWLEDGMENTS

The unusual title is intended as an informal mnemonic for the central idea; the theoretical construction and its limitations are presented in the conventional style of a modified-gravity manuscript. The author thanks the broader AeST literature for the covariant framework on which this proposal is based.

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